

Detecting departure and destination aerodromes from ADS-B

A methodology comparison against EUROCONTROL APDF ground truth

EUROCONTROL Performance Review Unit

2026-08-05

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Summary

The OPDI flight list assigns each flight a departure aerodrome (ADEP) and a destination aerodrome (ADES) from ADS-B alone. The production algorithm does this by testing the sign of a smoothed altitude trend near each candidate aerodrome and discarding every flight whose trend is ambiguous. It is accurate where it speaks — 98.25% on ADEP — but it stays silent on 45.22% of flights.

We compared seven methodologies against EUROCONTROL APDF ground truth. Three findings, in increasing order of consequence:

1. **Coverage is the problem, not accuracy.** Every trend-based method scores above 96% where it fires. None of them covers more than 58% of flights.
2. **The vertical-trend machinery earns nothing.** Attributing a cascade's answers to the rung that supplied them shows that the simplest possible rule — *the nearest aerodrome to the track's first and last sample* — is at least as accurate as every trend-based method **on that method's own flights**. A five-stage cascade built on those methods collapses exactly onto the one-line rule.
3. **What the trend methods actually supply is abstention, badly.** Replacing them with an explicit test on the endpoint — is it close to an aerodrome, and is it low over it? — beats the production algorithm on coverage and accuracy simultaneously: 58.79% of flights at 99.40% accuracy, against 54.78% at 98.25%.

 Interim — larger samples are being scored now

This is a working draft published while the benchmark is still running. Three-day samples in **2025-06** and **2024-06** are being scored on the OpenSky cluster and will replace every number below; the cleaning ablation, the per-aerodrome counts and the aerodrome-set sweep are implemented but not yet run.

The findings are reported now because the effects are large — several percentage points, not fractions of one — and because the central result is structural rather than numerical: it is about which signal carries the information, and that does not depend on the sample size.

 Scope of the results on this page

Every figure below is computed on a **single day, 2025-06-05** (32,404 ground-truth flights, 35.4 million state vectors). Three-day samples in 2025-06 and 2024-06 are being scored and will replace these numbers.

The sample is small in *days*, not in flights: three days is roughly 97,000 ground-truth flights, which is ample for fitting the two thresholds this study ends up recommending. The real exposure is representativeness — day of week, weather, seasonal traffic mix — which is why the second sample is a year apart rather than adjacent. Read the conclusions as what these days support.

1 Why this is hard

An ADS-B trajectory does not carry its origin or destination. Both must be inferred from where the aircraft was and what it was doing, and the inference fails in a specific, asymmetric way: reception near the ground is patchy. A receiver 50 NM from a regional aerodrome sees an aircraft in the cruise perfectly and never sees it below 2,000 ft. So the evidence that would settle the question — surface movement, the take-off roll, the flare — is exactly the evidence most often missing.

Three consequences shape everything below:

- **Tracks frequently begin or end in mid-air.** A track that starts in the cruise has no departure evidence at all, and any method that names an aerodrome for it is guessing.

- **Departure and destination are not symmetric.** Departure evidence is a climb from a known point; destination evidence is a descent to one. Coverage differs between the two, and so does accuracy.
- **Nearby aerodromes compete.** Within 30 NM of a major hub there are often several licensed aerodromes and, if heliports are included, dozens.

2 Data

2.1 Trajectories

State vectors are read from the OpenSky Network historical archive, filtered to the OPDI European bounding box and decimated to 5 s — byte-for-byte the same filter the production ingestion applies, so the benchmark evaluates the same population the pipeline ingests.

Tracks are built with OPDI’s own track-splitting algorithm, imported and called rather than reimplemented: `track_id` continuity with published data depends on it and it is frozen.

2.2 Aircraft selection

Only aeroplanes are in scope. Helicopters in particular defeat every aerodrome-based method — they depart from and arrive at places that are not in any aerodrome database — and they are not a rounding error: they are 9.8% of classified airframes in the OpenSky aircraft database.

Filtering uses the ICAO Doc 8643 aircraft class, keeping L (landplane) and A (amphibian) and dropping H (helicopter), G (gyrocopter) and T (tiltrotor). The field is present on ~73% of the database, so *unknown* handling is a real trade-off: the results here **exclude** unknowns, which protects precision at some cost to coverage. Two further layers — the ADS-B emitter category, which is the only field naming surface vehicles, and a behavioural filter requiring a sustained airborne period — are described under [Limitations](#) and not yet applied.

2.3 Ground truth

APDF `flights_tidy` provides ADEP, ADES and off-block time per flight. `AIRCRAFT_ADDRESS` is the ICAO 24-bit address, which is the join key to ADS-B; flights are matched on (`icao24`, `callsign`, `date`). 2.42% of those keys collide — same-day multi-leg rotations — and are resolved by proximity to the actual off-block time rather than left to chance.

3 How methods are scored

Ground truth is the denominator throughout. For a set of real flights G and a method that emits an answer on the subset $A \subseteq G$:

- **Coverage** = $|A|/|G|$ — the fraction of real flights the method answers at all. A flight whose trajectory was never seen counts as a miss.
- **Accuracy** = $|\text{correct}|/|A|$ — how often it is right *when it speaks*.
- **Overall** = $|\text{correct}|/|G|$ — coverage times accuracy; the fraction of real flights correctly labelled.

Reporting all three matters, because the production algorithm and its alternatives sit at opposite ends of the same trade: one is nearly always right and usually silent, the other nearly always speaks and is often wrong. Neither “accuracy” nor “coverage” alone can distinguish them.

i The coverage ceiling

No method exceeds ~73% coverage, because that is roughly the share of APDF flights with a usable ADS-B track in the sample at all. The remaining ~27% is a data-availability limit, not an algorithmic one, and no change to the naming rule can recover it.

4 The candidate methods

	Method	Signal
M0	<i>Current production</i>	Sign of a 5-sample rolling-mean altitude difference near each candidate aerodrome, requiring a margin of 4; ambiguous flights dropped
M1	<i>traffic library</i>	Nearest aerodrome to the track’s first sample (ADEP) and last sample (ADES). No altitude, no trend, no ground state
M2	<i>On-ground</i>	Aerodrome at the surface samples bracketing the airborne phase
M3	<i>Vertical rate</i>	Sign of the measured vert_rate near each candidate aerodrome
M5	<i>Lowest and nearest</i>	Aerodrome where the track is lowest and closest, split at the track midpoint
M6	<i>Cascade</i>	Take the most precise available answer, falling through when a rung is silent
M7	<i>Endpoint with abstention</i>	M1’s naming rule, but silent unless the endpoint is close to the aerodrome and low over it

M1 recreates `TakeoffAirportInference` / `LandingAirportInference` from the `traffic` library, which sorts by timestamp and calls `guess_airport` on the first and last row. M2 and M3 both use columns the production algorithm already selects and then never reads: `on_ground` and `vert_rate`.

Aerodromes come from `OurAirports`, restricted to large and medium airports. Including heliports is a swept parameter rather than a detail — `traffic`’s own documentation shows `infer_airport` returning a helipad instead of Amsterdam Schiphol when they are left in.

5 Results

Table 1: Coverage, accuracy and overall correctness by method, 2025-06-05, sorted by ADEP overall correctness.

Method	ADEP cov	ADEP acc	ADEP overall	ADES cov	ADES acc	ADES overall
M1_traffic	72.66%	87.54%	63.61%	71.25%	79.59%	56.71%
M6_cascade	74.40%	85.50%	63.61%	74.33%	76.31%	56.72%
M7_abstain	58.79%	99.40%	58.43%	50.23%	98.75%	49.61%
M5_min_alt	69.27%	82.16%	56.91%	68.99%	76.14%	52.53%
M3_vert_rate	57.42%	97.39%	55.92%	52.95%	97.06%	51.40%
M0_current	54.78%	98.25%	53.82%	52.66%	96.68%	50.91%
M2_on_ground	24.76%	96.57%	23.91%	28.11%	94.88%	26.67%

The shape of the table is the first finding. The production algorithm M0 is the **second most accurate** method tested and the **fifth** by overall correctness, because it declines to answer nearly half the flights. **traffic**’s one-line nearest-aerodrome rule, with no aviation logic in it whatsoever, gets more flights right in absolute terms than any trend-based method.

That is not an argument for adopting M1 as it stands — an 87.5% ADEP accuracy would be a serious regression in data quality for OPDI, which publishes a flight list that downstream users treat as authoritative. It is an argument that the trend logic is not where the value is.

6 The cascade, and why it fails

The natural response to the table above is a cascade: take M0’s answer when it speaks, fall through to M3, then M2, and use M1 only as a last resort. The strong signals keep their precision and M1 supplies coverage where they are silent.

It does not work, and the way it fails is instructive.

A cascade’s accuracy is *not* bounded below by its rungs’ accuracies, because each rung is scored only on the flights the rungs above it could not answer, which are the harder ones. So a cascade legitimately scoring below every method it is built from is not by itself evidence of a bug. What it must not do is score below its own fallback rung used alone — that means some rung is actively overwriting better answers.

To test that, every cascade answer is attributed to the rung that supplied it, and that rung is scored against a control on **the same flights**:

Table 2: ADEP attribution for the precision-ordered cascade M0 -> M3 -> M2 -> M1 -> M5, with M1 as the control. A negative lift means the rung overwrote a better answer.

Rung	Flights answered	Share of all	Its accuracy	M1 on the same flights	Lift
m0	17,751	54.78%	98.25%	99.55%	-1.30%
m3	907	2.80%	84.45%	92.50%	-8.05%
m2	240	0.74%	7.92%	47.08%	-39.17%
m1	4,685	14.46%	42.43%	42.43%	+0.00%
m5	526	1.62%	0.00%	0.00%	+0.00%

The lift is negative in every row. On the 17,751 flights M0 answers, M0 is 98.25% accurate and M1 is 99.55%. On M3's flights the gap is 8 points; on M2's, 39.

So the trend methods do not look precise because they reason well about trajectories. They look precise because they *fire on easy flights* — those with unambiguous low-altitude coverage near a single aerodrome — and on exactly those flights, “which aerodrome is nearest the endpoint” is already the right answer, slightly more often than the trend vote is. Where the trend vote differs, it is usually because a neighbouring aerodrome produced a stronger vote than the correct one.

Running the same cascade with M1 **first** confirms it: the remaining four rungs together contribute 564 additional flights, of which **none** are correct, and the cascade's score becomes indistinguishable from M1 alone.

💡 The reframing

The five methods differ in *when they are willing to answer*, not in *what they answer*. Their apparent precision is an abstention policy wearing a naming rule's clothes — and an expensive one, since it discards half the flights to buy accuracy it does not deliver. So the design question is not which signal should name the aerodrome. It is when to stay silent. And that is a property of the endpoint, not of the trajectory's shape.

7 Abstention on the endpoint

M7 keeps M1's naming rule and adds an explicit two-parameter test. An ADEP is emitted only if the track's first sample is within d NM of the nearest aerodrome and no more than h ft above that aerodrome's elevation; ADES uses the last sample. Height is measured above field elevation rather than above the ellipsoid, because a fixed 1,000 ft cut-off means nothing at an aerodrome sitting at 5,000 ft. A sample flagged `on_ground` satisfies the height test outright, since a missing barometric altitude on the surface is normal.

Two things are visible immediately.

The height threshold does nearly all the work. Moving the radius from 10 NM to 40 NM changes ADEP coverage by 0.2 percentage points. Moving the height cut from on-ground-only to 5,000 ft AGL moves it by 34 points. This makes physical sense: an aircraft below 5,000 ft AGL is in the terminal area of the aerodrome it is nearest to, and being 30 NM out at that height does not make it ambiguous. It also means the radius parameter, which is where the production algorithm spends its tuning effort, is close to irrelevant.

There is a region that dominates the production algorithm outright. The dotted lines mark M0's operating point. Every point up and to the right of their intersection beats it on coverage *and* accuracy at the same time:

Table 3: Selected operating points. Compare against M0 production: 54.78% / 98.25% for ADEP and 52.66% / 96.68% for ADES.

Within	Below	ADEP cov	ADEP acc	ADES cov	ADES acc
5 NM	1,000 ft AGL	44.52%	99.70%	46.60%	99.56%
10 NM	1,000 ft AGL	44.52%	99.70%	46.63%	99.52%
20 NM	1,000 ft AGL	44.54%	99.65%	46.76%	99.27%
40 NM	1,000 ft AGL	44.57%	99.58%	46.87%	99.07%

Within	Below	ADEP cov	ADEP acc	ADES cov	ADES acc
5 NM	2,000 ft AGL	53.22%	99.68%	47.61%	99.47%
10 NM	2,000 ft AGL	53.26%	99.66%	48.61%	99.40%
20 NM	2,000 ft AGL	53.30%	99.59%	48.75%	99.15%
40 NM	2,000 ft AGL	53.33%	99.53%	48.90%	98.90%
5 NM	5,000 ft AGL	56.73%	99.45%	47.89%	99.23%
10 NM	5,000 ft AGL	58.79%	99.40%	50.23%	98.75%
20 NM	5,000 ft AGL	58.91%	99.28%	51.74%	98.33%
40 NM	5,000 ft AGL	58.99%	99.18%	52.02%	97.97%

At 10 NM and 5,000 ft AGL the rule answers 58.79% of ADEPs at 99.40% accuracy — four points more coverage than production, at higher accuracy. Tightening to 2 NM and 1,000 ft trades coverage down to 43% for 99.69% accuracy, which is essentially error-free labelling. The parameters give OPDI a dial where it currently has a fixed point, and every position on the dial between 1,000 and 5,000 ft is above 99% accurate.

ADES is consistently weaker than ADEP at the same thresholds — about 6 points less coverage at 5,000 ft AGL. Arrivals are harder: a descent can be interrupted by a go-around or a diversion, and the last recorded sample of a track is more often a loss of reception than a landing.

8 Recommendation

On the evidence of one day, OPDI should replace the altitude-trend vote with endpoint proximity plus explicit abstention. This is a smaller algorithm than the one it replaces — no smoothing window, no margin parameter, no per-aerodrome vote — and it is strictly better on both axes.

The two parameters should be published alongside the flight list, and the abstention reason retained per flight, so that downstream users can distinguish “no departure aerodrome” from “departure aerodrome not determinable”. The current algorithm collapses those two cases into a null, which is a loss of information that costs nothing to preserve.

This should not be implemented until the full-month results are in.

9 Limitations

- **One day, becoming three plus three.** Everything here rests on 2025-06-05. Three-day samples in 2025-06 and 2024-06 are being scored; fitting the thresholds on one and validating on the other, a year apart, is the only honest way to report a tuned parameter. Full-month sampling was considered and dropped: it costs ~12 hours of cluster time and ~100 GB to reduce a sampling error that is already small relative to the effect sizes here — the gap between M7 and the production algorithm is four percentage points of coverage and one of accuracy, not a fraction of a point.
- **No trajectory cleaning.** The state vectors are raw apart from the bounding box and decimation. OPDI’s cleaning stage has not been applied, and whether it helps here is an open question rather than a formality: the stale-broadcast filter nulls repeated positions, which is precisely what a stationary aircraft at a gate transmits. It could remove the very samples the endpoint rule depends on. This will be measured as an ablation, not assumed.

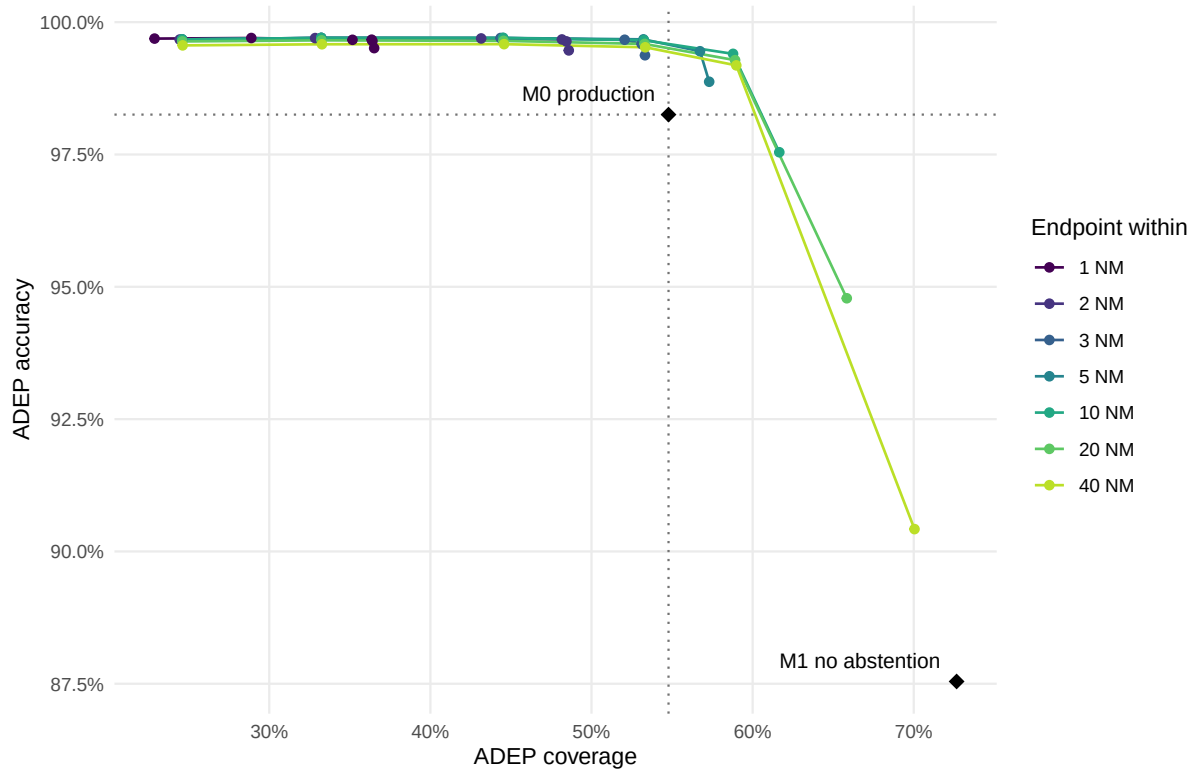


Figure 1: Coverage/accuracy frontier for endpoint abstinence, ADEP, 2025-06-05. Each line is one distance threshold; points along it are height thresholds, relaxing left to right from on-ground-only through 5,000 ft AGL to unrestricted. Accuracy holds above 99% across a long plateau and then falls off a cliff once the height test is dropped altogether. The dotted lines mark the production algorithm M0; the region up and to the right of their intersection beats it on both axes at once.

- **Aircraft filtering is one layer of three.** The ADS-B emitter category — the only field that names surface vehicles — is dropped by the reference pipeline stage and would need to be retained. A behavioural filter requiring a sustained airborne period is database-independent and would catch the 27% of airframes with no recorded class at all.
- **Per-aerodrome counts are not yet reported.** Comparing arrival and departure counts per aerodrome against APDF is a stricter test than flight-level accuracy and is only meaningful at the 90 aerodromes APDF covers. It is not in this version.
- **Aerodrome set is fixed.** All results use large and medium airports. Sweeping small airports and heliports is implemented but not yet reported.

10 Reproducing

Everything is in the `opdi` repository under `benchmarks/`, and every dataset written is registered in `benchmarks/DATASETS.md`.

```
# 1. Sample state vectors from the OpenSky archive into S3
python benchmarks/osn_sample.py 2025-06-05 2025-06-08

# 2. Score every method
python benchmarks/adept_ades.py --days 2025-06-05 --months 202506 --methods all
```

```
# 3. Attribute the cascade's answers, with M1 as the control
python benchmarks/adeq_ades.py --days 2025-06-05 --months 202506 \
    --methods M6_cascade --diagnose-cascade --control m1

# 4. Sweep the abstention grid
python benchmarks/adeq_ades.py --days 2025-06-05 --months 202506 \
    --methods none --sweep-abstain

# 5. Refresh this page's CSV cache
python benchmarks/export_results.py
```

Steps 1–4 run on the OpenSky Kubernetes cluster and need credentials. Step 5 needs S3 read access. Rendering this page needs neither.